Trails — Product Requirements Document

Version 0.1.0 · Private Scans

Status: Proposed implementation baseline
Platform: Native iOS
Starting point: Fork of Extensa's existing capture and spatial-viewing implementation
Release focus: Capture, privately preserve, review, and revisit indoor trails

1. Product definition

Trails lets people capture their movement through a place and preserve the observations, memories, and moments they associate with it.

Its long-term purpose is to connect community, identity, and living heritage:

> A place's story includes the people who passed through it and what they chose to leave behind.

Version 0.1.0 establishes the foundation: **private** indoor scans and trails stored on the user's device.

A person can capture a space, record a path, add meaningful checkpoints, and return to that experience later. Public contributions, community discovery, and Logos publishing build on this foundation in subsequent releases.

1.1 What this release is

- A working indoor capture and trail-review application.
- A personal collection of spatial journeys.
- An authored experience: each trail has a creator and optional personal context.
- A local-first application with no account or network dependency for its core workflow.

1.2 What this release is not

- A production indoor-navigation system.
- A verified historical archive.
- A social network.
- An outdoor terrain scanner.

- A public place directory.
- A complete LP-0018 implementation.
- A promise of permanent, encrypted, recoverable cloud storage.

2. Product principles

Principle	Requirement
People give places meaning	Trails support a personal introduction and place-linked moments, not just geometry.
Capture first, organise afterward	Starting a recording must not require a lengthy form.
Private by default	No capture or media is uploaded by the application in this release.
Preserve original observations	User-written content and recorded samples remain distinct from calculated presentation data.
Reuse working technology	Retain Extensa's functioning capture, plan, and footsteps components wherever possible.
Honest spatial representation	Do not imply accurate live positioning or automatic alignment between unrelated scans.
Design for the future without building it now	Stable identifiers and a versioned local format must permit later export, sharing, and publishing.

3. Goals and success criteria

3.1 Primary goal

A user can capture an indoor trail, save it, close the application, reopen it, and explore the saved space, path, and moments.

3.2 Release success criteria

1. A supported device completes a real indoor capture.
2. The resulting trail is stored locally without an account or internet connection.
3. Saved geometry, path, metadata, and moments survive application restart.
4. The user can inspect the saved trail in 3D and 2D.
5. The user can edit local metadata and delete the trail.
6. The library clearly distinguishes unfinished recordings from saved trails.
7. The interface never suggests that local content has been published or backed up.
8. A second recording does not overwrite or corrupt the first.
9. Capture interruption produces an honest recovery or failure state.
10. The complete loop is demonstrated using an actual capture, not only fixtures.

3.3 Demonstration narrative

> “I captured this place, left a few moments along my journey, and saved it privately. Now I can return to the space and see it through that recorded perspective.”

4. Users and use cases

4.1 Primary user: contributor

A person who wants to preserve a journey through an indoor place.

Examples:

- A resident documenting a meaningful corner of Dhun.
- A visitor recording details they want to remember.
- A person explaining a space through labels and observations.

4.2 Secondary user: returning explorer

The same person revisiting a saved trail.

They want to:

- Recognise the space.
- Inspect the route they recorded.
- Read their observations.
- Review individual moments.
- Prepare the contribution for possible sharing later.

4.3 Later-release users

Not supported as networked roles in v0.1.0:

- Friends receiving shared trails.
- Public contributors.
- People searching community place collections.
- Independent Logos consumers.

There is no administrator or place-approval role in the product model.

5. Scope

5.1 Required for v0.1.0

Area	Included
Local identity	Display name; optional introduction
Capture	Existing supported indoor scanning workflow
Path recording	Timestamped positions from the existing spatial-tracking implementation
Moments	Named checkpoints with optional text notes
Spatial review	3D capture, 2D plan, recorded path
Trail editing	Title, place label, floor, introduction, moments
Local library	Saved trails, drafts, search, sorting
Persistence	Versioned metadata and locally stored assets
Privacy	No automatic uploads or required online services
Recovery	Incremental metadata/path saving and clear interrupted-session handling
Deletion	Delete local trail and its owned assets
Diagnostics	Clear unsupported-device, permission, storage, and processing errors

5.2 Optional only after the required flow works

- Cover selection from an existing image or preview.
- Photos attached to moments.
- Short audio notes attached to moments.
- Remembered 2D/3D viewing preference.
- Small path-reveal animation.
- Developer-only export for debugging and backup testing.

Do not delay the core loop for optional media.

5.3 Explicitly excluded

- Public or unlisted publishing.
- Shareable web links and QR distribution.
- Logos Storage or blockchain integration.
- On-chain registries.
- OSM map display or export.
- Global place search.
- Cloud accounts, authentication, or synchronisation.

- AI transcription, storytelling, or navigation.
- Turn-by-turn directions.
- Automatic pathfinding or rerouting.
- Background capture.
- Cross-device localisation.
- Automatic alignment of independently recorded scans.
- Likes, comments, ratings, followers, rewards, or territory ownership.

6. Platform and technical boundaries

6.1 Platform

Native iOS application, using the existing Extensa-supported framework versions and deployment target.

Before implementation begins, the repository must record:

- Minimum iOS version.
- Required device capabilities.
- Devices actually tested.
- Whether non-capture viewing is supported on devices that cannot scan.

Do not choose a deployment target from a design mockup. Derive it from the working capture implementation.

6.2 Existing implementation assumption

The user reports that Extensa already supports:

- Indoor geometry capture.
- A simplified 3D representation.
- A 2D plan.
- Footsteps or a recorded path.

These are starting assumptions, not acceptance evidence. The first engineering task is to run the existing code and identify exactly which outputs can be saved and restored.

6.3 Capture versus tracking

Keep these responsibilities separate:

Responsibility	Output
Indoor capture	Geometry and recognised spatial elements

Responsibility	Output
Spatial tracking	Time-ordered positions used for the recorded path
Trail authoring	Labels, notes, moments, and introduction
Review rendering	Views generated from stored outputs

Do not assume the scan export automatically contains the complete recorded walking path.

6.4 Coordinate system

- Preserve the capture session's local coordinate frame.
- Store its convention and units in the file schema.
- Store path samples and moments relative to the same frame.
- Do not label local XYZ positions as geographic coordinates.
- Do not convert local paths to GeoJSON longitude/latitude without an explicit geographic transformation.
- Geographic anchoring is not required for this release.

7. Information architecture

7.1 Primary navigation

Three destinations:

Destination	Purpose
My Trails	Search, open, and manage local recordings
Record	Begin a new capture
You	Edit the local contributor identity

Implementation may use a central Record action rather than a full tab if it better fits the existing application.

There must be no empty public Explore tab implying unavailable functionality.

7.2 Core screens

1. Welcome and local identity.
2. My Trails.
3. Capture preparation.
4. Active capture.
5. Add moment.
6. Capture processing.

7. Review and edit.
8. Saved trail detail.
9. Full-screen spatial exploration.
10. Profile and local-data settings.

8. User journeys

8.1 First launch

```
Open app
→ Read short product introduction
→ Set display name or skip
→ Open My Trails
→ Start first recording
```

Requirements:

- No email or password.
- Interests are optional and may be omitted entirely.
- Users may begin under a default local identity.
- Camera or microphone permissions are requested contextually.
- State that captures stay on the device and are not automatically backed up by Trails.

8.2 Record a trail

```
Tap Record
→ Check device support and permissions
→ Begin capture
→ Move through the indoor space
→ Add moments
→ Finish
→ Process capture
→ Review
→ Save locally
```

Requirements:

- Generate an ID and draft record before capture begins.
- Do not require a title before recording.
- Use the existing capture guidance.
- Show an unmistakable recording state.
- Show tracking interruption separately from successful tracking.
- Require explicit confirmation before discarding an active recording.

8.3 Add a moment

```
Tap Add moment
→ Capture current valid spatial position
→ Enter label and optional note
→ Save
→ Return to recording
```

If no valid position is available:

- Explain that the moment cannot yet be spatially placed.
- Offer Retry.
- Alternatively save it as an explicitly unplaced note.

Never silently anchor a moment at an arbitrary origin.

8.4 Save and reopen

```
Finish capture
→ Finalise and persist available assets
→ Edit optional metadata
→ Return to library
→ Close app
→ Reopen
→ Open saved trail
```

Finishing capture and editing metadata are separate operations. Closing the metadata screen must not discard a successfully saved capture.

8.5 Review a trail

```
Open trail
→ See cover or spatial preview
→ Read title and introduction
→ Explore 3D or Plan
→ Select moment
→ Highlight corresponding point
→ Read note
```

8.6 Delete a local trail

```
Open trail menu
→ Delete
→ Confirm
→ Remove trail and its owned assets
→ Return to library
```

No published-content restrictions apply because v0.1.0 contains only private local records.

9. Functional requirements

9.1 Local identity

ID	Requirement
ID-01	Create a stable local creator ID on first use.
ID-02	Permit a display name without account creation.
ID-03	Permit editing display name and introduction.
ID-04	Do not imply that the identity is verified or globally unique.
ID-05	Display creator attribution in trail detail and moments.

Recommended default attribution: “A trail by you” until a name is supplied.

9.2 Capture

ID	Requirement
CAP-01	Check supported hardware before starting.
CAP-02	Request required permissions with understandable explanations.
CAP-03	Create a recoverable draft before opening the capture session.
CAP-04	Present the existing 3D capture view and tracking guidance.
CAP-05	Show the existing 2D plan/footsteps where available.
CAP-06	Persist path samples incrementally.
CAP-07	Support Finish and Cancel.
CAP-08	Stop recording when capture ends.
CAP-09	Surface interruptions and degraded tracking.
CAP-10	Do not guarantee continuous recording while backgrounded or locked.

Pause/resume is included only if Extensa already supports it without breaking the coordinate frame. Otherwise omit it from v0.1.0.

9.3 Moments

ID	Requirement
MOM-01	Create a moment during recording.
MOM-02	Require a nonempty label.
MOM-03	Permit a text note.
MOM-04	Store creation time and associated sample/session reference.

ID	Requirement
MOM-05	Store a valid spatial position or explicit unplaced status.
MOM-06	Permit label/note edits after recording.
MOM-07	Permit deleting a moment.
MOM-08	Preserve recorded order; list reordering is optional.

9.4 Review and rendering

ID	Requirement
REV-01	Open a saved capture without rescanning.
REV-02	Provide 3D and 2D plan views where supported by stored data.
REV-03	Render the saved path in the correct local frame.
REV-04	Display selectable moment markers.
REV-05	Highlight a selected moment and show its note.
REV-06	Provide Fit scene/Recenter view.
REV-07	Offer Previous/Next moment controls.
REV-08	Label the view as a saved recording, not live navigation.
REV-09	Do not request camera access merely to inspect a saved model unless the chosen viewing mode requires it.

9.5 Library

ID	Requirement
LIB-01	List local trails newest first by default.
LIB-02	Support search across title and place label.
LIB-03	Show title, thumbnail, place, date, and state.
LIB-04	Distinguish Draft, Processing, Saved, Interrupted, and Failed.
LIB-05	Provide a useful empty state and Record action.
LIB-06	Open records without loading every full 3D asset into memory.
LIB-07	Permit deletion with confirmation.

9.6 Metadata

Editable fields:

- Title.
- Place label.
- Optional floor.

- Introduction.
- Moment labels and notes.
- Optional cover selection if implemented.

Default title: “Untitled trail.”

A place label is local metadata in this release, not a global place-registry entry.

10. UI requirements

10.1 Visual identity

Trails should communicate:

- Human authorship.
- Spatial exploration.
- Quiet curiosity.
- Connection to a real place.

Suggested palette:

- Forest green.
- Warm limestone.
- Terracotta for the active path.
- Dark ink text.
- Restrained brass accents.

10.2 Screen hierarchy

My Trails

- Search.
- Compact local trail cards.
- Clear state labels.
- Prominent Record action.

Active capture

- Spatial viewport dominates.
- Minimal controls.
- Add moment and Finish remain easy to reach.
- No social notifications or celebratory overlays.

Review

- Spatial preview first.
- Metadata below.
- Moments visible and editable.
- Clear “Saved on this device” state.

Explore saved trail

- Full-screen spatial content.
- 3D/Plan toggle.
- Moment sheet.
- Previous/Next.
- Minimal persistent chrome.

10.3 Accessibility

- Support Dynamic Type for text-based screens.
- Label controls for VoiceOver.
- Do not use colour alone to indicate selection.
- Respect reduced-motion preferences.
- Provide text alternatives for key moment information.
- Keep destructive actions separate from primary actions.
- Maintain adequate contrast and touch-target sizes.

Spatial capture itself is not claimed to be an accessible navigation aid.

11. Local storage design

11.1 Storage layout

Proposed application-private layout:

```
Application Support/
  Trails/
    library-index
    profiles/
      local-profile.json
    trails/
      <trail-id>/
        manifest.json
        capture/
          ...
        tracking/
          samples
```

```
        moments.json
        previews/
          cover
          plan
        media/
          ...
```

Exact file types should follow the existing implementation where practical.

11.2 Storage requirements

- Store private files outside public/shared locations by default.
- Save manifests atomically.
- Avoid storing large binary assets directly in a lightweight metadata index.
- Maintain one authoritative record of each asset's ownership.
- Ensure deletion does not remove assets belonging to other trails.
- Rebuild or repair the library index from manifests where feasible.
- Keep temporary processing assets separate from final assets.
- Include schema versions.
- Document the OS backup policy.

"On-device" does not necessarily mean "excluded from operating-system backups." Decide whether capture assets are backup-eligible and describe that behaviour accurately.

11.3 Security wording

Allowed:

> "Saved privately on this device. Trails does not upload this capture."

Not allowed without implementation evidence:

- "End-to-end encrypted."
- "Only you can ever access it."
- "Permanently preserved."
- "Recoverable if your phone is lost."
- "Never included in device backups."

Use platform file-protection mechanisms appropriately and document the chosen protection class.

12. Data model

12.1 Trail

```
Trail
  id
  schemaVersion
  creatorId
  title
  placeLabel?
  floorLabel?
  introduction?
  createdAt
  updatedAt
  lifecycleState
  visibility = localPrivate
  captureAssetReferences
  pathAssetReference?
  coordinateFrame
  momentIds[]
  previewReferences
```

12.2 Coordinate frame

```
CoordinateFrame
  id
  units
  axisConvention
  sessionReference
  geographicAlignment = none
```

12.3 Path sample

```
PathSample
  timestamp
  positionX
  positionY
  positionZ
  trackingState
  sessionSegmentId
```

Persist orientation only if existing rendering or future requirements need it.

12.4 Moment

```
Moment
  id
  trailId
  label
```

```
  note?
  createdAt
  order
  spatialPosition?
  placementStatus
  pathSampleReference?
  mediaReferences[]
```

12.5 Local profile

```
LocalProfile
  id
  displayName
  introduction?
  portraitReference?
  interests[]
```

No password, email, remote identity token, or public username registry is required.

13. Lifecycle and recovery

13.1 State machine

```
Draft
  → Recording
  → Processing
  → Saved

Recording → Interrupted
Processing → Failed
Interrupted → Recoverable draft or explicit failure
Failed → Retry processing, if retained assets permit
```

Deletion is available for all inactive local states.

13.2 Recovery rules

- Metadata and moments should survive interruption if already saved.
- Path samples should be persisted periodically during capture.
- Final room geometry may not be recoverable after an abrupt interruption; do not promise otherwise.
- Retain usable partial outputs when safe and meaningful.
- Never label an incomplete capture “Saved successfully” without checking its required assets.
- On launch, inspect records left in Recording or Processing and reconcile their state.

14. Application architecture

Preserve existing Extensa modules and add a thin Trails domain layer.

```
SwiftUI application
  ├── Library UI
  ├── Capture UI
  ├── Review UI
  └── Profile UI

Trail domain
  ├── TrailRepository
  ├── CaptureCoordinator
  ├── PathRecorder
  ├── MomentService
  └── PreviewGenerator

Existing spatial implementation
  ├── Room capture
  ├── Tracking
  ├── 3D renderer
  └── 2D plan renderer

Local persistence
  ├── Metadata index
  ├── Versioned manifests
  └── Asset storage
```

Architectural rules

1. Views do not directly own persistence logic.
2. Capture callbacks do not perform heavy synchronous disk work on the UI thread.
3. Original capture outputs remain separate from thumbnails and derived previews.
4. Future sharing must consume a saved trail package, not reach into a live capture session.
5. Do not introduce a backend merely to support local identity or local storage.
6. Avoid rewriting Extensa's working rendering pipeline for visual consistency alone.

15. Performance and quality targets

These are proposed release targets to verify on the actual demo device.

Area	Target
Library	Usable with at least 100 lightweight local records
Asset loading	Full captures loaded on demand
UI	No obvious blocking during routine metadata edits
Capture	Persistence must not noticeably disrupt tracking

Area	Target
Save feedback	Show immediate progress/status for processing operations
Durability	Successful save survives process restart
Resource use	No unbounded growth in in-memory path or preview objects
Offline	Entire required user journey completes without network connectivity

Test using realistic capture sizes before claiming specific loading-time guarantees.

16. Testing requirements

16.1 Unit tests

- Trail creation and defaults.
- Metadata serialisation/deserialisation.
- Moment creation and edits.
- Coordinate-frame references.
- State transitions.
- Manifest version handling.
- Search and sorting.
- Asset deletion ownership.
- Invalid or missing asset handling.

16.2 Integration tests

- Save and reload a real capture.
- Save and reload path samples.
- Preserve moment positions across reopening.
- Update metadata without changing capture assets.
- Delete one trail without affecting another.
- Recover an interrupted draft.
- Open a record with a missing optional preview.
- Fail gracefully when a required asset is unavailable.

16.3 Manual device tests

1. Fresh installation.
2. Unsupported-device handling where testable.
3. Camera permission denied.

4. First capture.
5. Multiple captures.
6. Capture with several moments.
7. Application backgrounded during recording.
8. Application terminated after a successful save.
9. Application reopened without internet.
10. Large capture opened and closed repeatedly.
11. Local deletion.
12. VoiceOver and larger text on non-spatial screens.

17. Acceptance checklist

Core loop

- [ ] Start a trail without creating an online account.
- [ ] Capture an actual indoor space.
- [ ] Record a path in the same coordinate frame.
- [ ] Add at least three labelled moments.
- [ ] Finish and persist available outputs.
- [ ] Reopen the trail after restarting the app.
- [ ] See the correct geometry, path, and moments.

Review

- [ ] Switch between 3D and Plan.
- [ ] Select a moment and read its note.
- [ ] Edit local title and introduction.
- [ ] Return to the library without losing changes.

Privacy and storage

- [ ] No application upload occurs.
- [ ] No analytics dependency exists.
- [ ] The UI clearly says content is local.
- [ ] Backup behaviour is documented.
- [ ] Local deletion removes the trail's owned files.

Honesty

- [ ] Saved replay is not presented as live navigation.
- [ ] No public link is displayed.
- [ ] No unimplemented Logos integration is claimed.
- [ ] No automatic alignment of independent scans is implied.
- [ ] Sample content is identifiable as sample content.

18. Implementation sequence

Phase A — Verify the fork

- Run Extensa unchanged.
- Confirm device support.
- Identify capture, path, plan, and render outputs.
- Confirm what can already be saved and reopened.
- Create the Trails application identity.

Exit condition: existing functionality still works.

Phase B — Local trail persistence

- Stable IDs.
- Manifest and asset layout.
- Local library.
- Save/reopen.
- Clear lifecycle states.

Exit condition: one real capture survives restart.

Phase C — Authorship and moments

- Local profile.
- Title and introduction.
- Checkpoint labels and notes.
- Moment selection in review.

Exit condition: the recording communicates a person’s perspective, not only geometry.

Phase D — Review and polish

- 3D/Plan switch.
- Path visibility.
- Moment navigation.
- Search and compact cards.
- Errors, empty states, deletion.
- Device testing.

Exit condition: the required acceptance checklist passes.

Phase E — Optional enhancements

Only after the core loop:

- Photo/audio attachments.
- Better previews.
- Path-reveal animation.
- Developer export.

19. Subsequent releases

Release direction	Capability
Portability	Export/import versioned trail packages
Private sharing	Encrypted package handling and explicit recipient/access model
Logos integration	Publish and retrieve packages through a tested storage adapter
Public contribution	Place-linked public trails and creator profiles
Discovery	Geographic overview, search, place collections
Spatial interoperability	Explicit alignment and reviewed export approaches
Assistance	Transcription, interpretation, and route assistance
Navigation	Separately tested live positioning and guidance

The future public model remains open contribution: no place owner has exclusive publishing rights. Public immutability and directory visibility must be specified when publishing is implemented; they do not restrict local drafts in v0.1.0.

20. Definition of done

Trails v0.1.0 is complete when a person can:

> Capture an indoor space, record their trail, attach meaningful observations, save it privately, and return later to explore the same spatial journey.

The release succeeds by making that loop reliable and expressive. Public discovery, sharing, AI, and Logos integration must not delay or obscure that foundation.